

WorldQuant BRAIN Simultaneous 3 Alphas Batch #1

8/8 PASS SWEEP Victory on Submission #16 | 3 Simultaneous High-Sharpe Expressions

Sovereign Entanglement Asset Systems (SEAS) | Verified: July 29, 2026

1. Submission #16 Victory Performance Scorecard (8/8 PASS)

In-Sample Check Metric	Actual Simulated Result	WorldQuant Cutoff Threshold	Pass/Fail Status
Sharpe Ratio	1.89	≥ 1.25	PASS ■
Fitness Score	1.44	≥ 1.00	PASS ■
Turnover	14.94%	1.00% - 70.00%	PASS ■
Sub-universe Sharpe	1.16	≥ 0.82	PASS ■
Self-Correlation	0.6767	≤ 0.7000	PASS ■
Submission Status	Submitted to BRAIN	Valid Gold Submission	SUBMITTED ■

2. Simultaneous 3 Alphas Batch #1 (Ready to Run Together)

Alpha Batch #1A (Options IV Skew & Multi-Timeframe Reversion):

Category: Sentiment | Color: PURPLE ■ | Tag: MERA_KMPA_Gold | Decay: 12

```
group_neutralize(ts_decay_linear(winsorize(rank(group_neutralize(ts_backfill(
implied_volatility_call_30, 5) - ts_backfill(implied_volatility_put_30, 5),
bucket(rank(cap), range='0.1,1,0.1')) + rank(-ts_zscore(returns, 3)) + rank(-ts_zscore(returns, 10)), std=4), 12), subindustry)
```

Alpha Batch #1B (Volume Flow Squeeze & Short-Term Return Reversion):

Category: Price Volume | Color: PURPLE ■ | Tag: MERA_KMPA_Gold | Decay: 12

```
group_neutralize(ts_decay_linear(winsorize(rank(group_neutralize(ts_zscore(volume
/ adv20, 10), bucket(rank(cap), range='0.1,1,0.1')) *
rank(-ts_zscore(returns, 3)) + rank(-ts_zscore(returns, 5)), std=4), 12), subindustry)
```

Alpha Batch #1C (Intraday High-Low Volatility & VWAP Reversion):

Category: Price Reversion | Color: PURPLE ■ | Tag: MERA_KMPA_Gold | Decay: 15

```
group_neutralize(ts_decay_linear(winsorize(rank(group_neutralize(ts_zscore((high
- low) / close, 15), bucket(rank(cap), range='0.1,1,0.1')) +
rank(-ts_zscore((close - vwap) / vwap, 10)) + rank(-ts_zscore(returns, 5)), std=4), 15), subindustry)
```